

Vanishing gradient problem:-

1) $0.1 \times 0.1 \times 0.1 = \boxed{0.0001} \rightarrow \text{VGP}$

2) Deep NN

3) Sigmoid/tanh $\rightarrow \Delta \cdot \text{func} \rightarrow$

$\rightarrow$ In case when layers becomes too high

Backprop converge not & model didn't trained

$\rightarrow$ Simply that's a big problem as when model not trains that's a problem.

How to recognize:-

1) Focus on loss

no changes $\rightarrow$ VGP

2) weights $\rightarrow$ plot graph

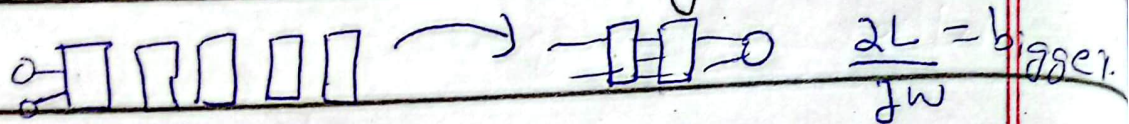
$\square$ epochs

Date: _____

Day: _____

How to handle Va. Gd. Pb:-

1) Reduce model complexity:-

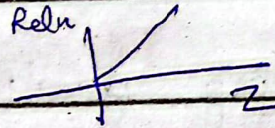


Not too applicable as we don't

find out complex pattern.

2) Using relu act. func:-

$\max(0, z)$



-1000 - 1000

0 - 1000 squish

output is 0, 1 so when u multiply

2×2 so it doesn't b/m smaller

& problem solved.

3) Proper weight initialization:-

4) Batch norm $\rightarrow$ layer $\rightarrow$

5) Using residual network.

Date: _____

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Exploiting Gr. prblm :- (KNN)

Complete opposite of va. Gr. pb.

→ makes no much larger.

10, 10, 10 → 100000

$$\frac{\partial L}{\partial w} = d_1 \times d_2 \times d_3$$

$$w_{\eta} = w_0 \eta \frac{\partial L}{\partial w} \rightarrow 1000$$

$$1 - 100 = 99 \dots$$

§ loss not reduced.

Memoization:-

an optimization technique used to speed up computer programs by storing results of expensive function calls & returning cached result when input occur again.

def fib(n):

if n == 1 or n == 0

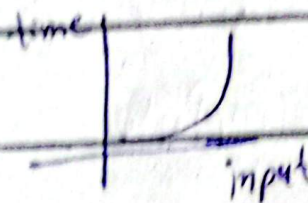
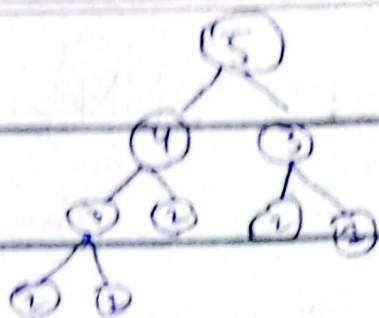
return 1

else:

return fib(n-1) + fib(n-2)

ate: _____

Day: _____



its a trick u used to reduce time
by sacrificing space.

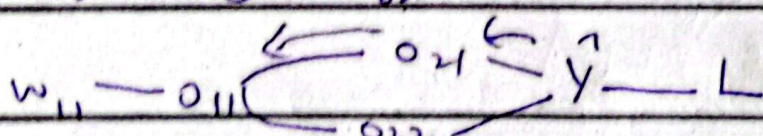
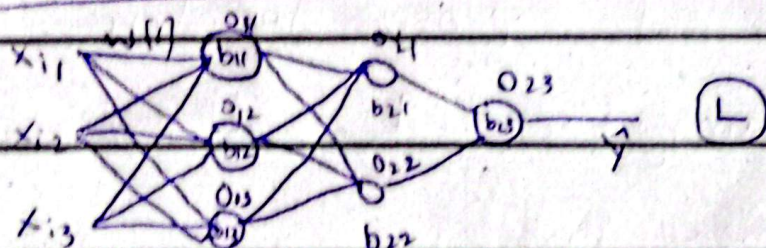
→ In case of 2 hidden layers we have
to find a bundle of derivatives for 1 term.
so memoization helps to store those answer
for future use as it takes alot time else.

Backprop → chainrule (diff) + memoization



maths + Cs.

keras + TF



$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial y} \left[\frac{\partial y}{\partial o_{21}} \times \frac{\partial o_{21}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial w_{11}} + \frac{\partial y}{\partial o_{22}} \times \frac{\partial o_{22}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial w_{11}} \right]$$